Take-Home Part II

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Question 1

A major difficulty in what we did previously was determining a good covariance matrix to use in the construction of confidence intervals based on asymptotic normality, regardless of whether that asymptotic normality was based on maximum likelihood or maximum composite likelihood estimation. One possible alternative to the direct use of asymptotic results is to compute intervals directly using parametric bootstrap methods.

(a)

Determine 95% confidence intervals for the five parameters of the model through the application of percentile bootstrap intervals for maximum likelihood estimation.

Methodology. We construct 95% percentile bootstrap confidence intervals for each of the five model parameters $(\alpha, \beta, \gamma, \theta, \sigma^2)$ following the development in Section 10.3. The percentile interval is derived by assuming there exists some monotone symmetrizing transformation $\varphi = W(\hat{\theta}_n)$ of the estimator; crucially, as noted following expression (10.15), the interval can be computed *without ever determining what that transformation is*. The result is that the $(1-\alpha)100\%$ percentile bootstrap interval for a scalar parameter takes the form

$$\left((\theta_n^*)_{[(M+1)\alpha/2]},\ (\theta_n^*)_{[(M+1)(1-\alpha/2)]}\right) \tag{10.15}$$

where $(\theta_n^*)_{[k]}$ denotes the kth order statistic of the M bootstrap estimates. That is, the interval endpoints are simply the empirical $\alpha/2$ and $1-\alpha/2$ quantiles of the bootstrap distribution of $\theta_{n,m}^*$.

Simulation from the fitted model. We take $M = 999$ so that $(M+1)\cdot 0.025 = 25$ is an integer, satisfying the integrality condition noted after expression (10.10). For each bootstrap replicate $m = 1, \ldots, M$, we simulate $\mathbf{y}_m^* = (y_1^*, \ldots, y_n^*)^T$ from the fitted model $F(\mathbf{y} \mid \hat{\theta}_n)$ — the model estimated at the MLE from the observed data, consistent with the general simulation framework described in Section 10.1. Because the model includes AR(1) errors, simulation requires drawing $\varepsilon_1^* \sim N\Big(0,\, \hat{\sigma}^2/(1-\hat{\theta}^2)\Big)$ from the stationary distribution and then propagating $\varepsilon_i^* = \hat{\theta}\,\varepsilon_{i-1}^* + \delta_i$ with $\delta_i \overset{iid}{\sim} N(0, \hat{\sigma}^2)$ for $i = 2, \ldots, n$. The simulated response is then $y_i^* = \hat{\mu}_i + \varepsilon_i^*$ where $\hat{\mu}_i = \hat{\alpha}\, x_i^{\hat{\beta} x_i^{-\hat{\gamma}}}$.

Re-estimation. Each bootstrap dataset $\mathbf{y}_m^*$ is re-fit by maximizing the same log-likelihood as in Q3, using the original MLE $\hat{\theta}_n$ as the starting point for BFGS optimization. This yields bootstrap estimates $\theta_{n,m}^*$ for $m = 1, \ldots, M$ in the notation of Section 10.1. Any replicate for which the optimizer fails to converge is discarded; we do not re-run to replace failed replicates, as repeatedly resampling until a convergence target is met would selectively exclude regions of the likelihood surface where optimization is difficult, potentially biasing the bootstrap distribution.

Interval construction. Of the $M = 999$ replicates, $M_{\text{eff}} = 988$ converged successfully and were retained (using `set.seed(42)`). Following expression (10.15), the 95% percentile interval for each parameter is the empirical 2.5% and 97.5% quantiles of the $M_{\text{eff}} = 988$ bootstrap estimates, computed via interpolation rather than exact order-statistic indexing since $(M_{\text{eff}} + 1) \cdot 0.025 = 24.725$ is no longer an integer. As Section 10.5 notes, discarding non-converging replicates conditions uncertainty estimation on estimable datasets, which

may modestly understate uncertainty; with only 11 failures out of 999 this effect is expected to be negligible. No discrepancy function $h(\theta, \hat{\theta}_n)$ needs to be formed or inverted — unlike the basic interval (10.10) or studentized interval (10.12) — because the percentile method works directly on the empirical distribution of $\theta^*_{n,m}$.

It is worth noting the distinction from what Section 10.1 cautions against: we are *not* treating the empirical distribution of $\{\theta^*_{n,m}\}$ as a direct approximation to the sampling distribution of $\hat{\theta}_n$. Rather, the percentile method's validity rests on the existence of a transformation under which the basic bootstrap interval and the percentile interval coincide, as shown in expressions (10.13) through (10.15). The asymptotic-normal intervals from Q3 are included in the table for comparison; the bootstrap offers an alternative that does not require directly inverting the observed information matrix for standard errors. However, this by no means eliminates the concerns of Part I regarding whether the relevant asymptotics hold. That being said, we nonetheless proceed.

Table 1: 95% Percentile Bootstrap and Asymptotic-Normal CIs

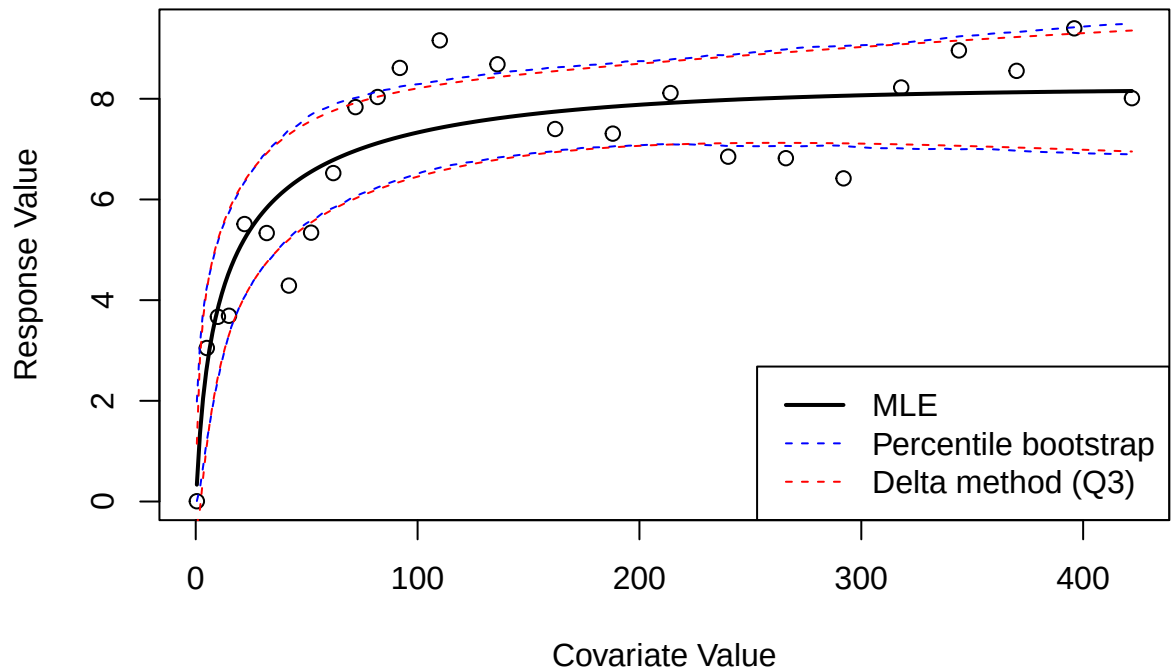
Parameter	MLE	Boot 2.5%	Boot 97.5%	AN 2.5%	AN 97.5%
α	0.7395	0.0675	2.5107	-0.4834	1.9625
β	1.0283	0.3085	2.5010	0.0275	2.0291
γ	0.1574	0.0552	0.2099	0.0929	0.2219
θ	0.5598	-0.1289	0.6488	0.2203	0.8993
σ^2	0.6250	0.2633	0.8790	0.2785	0.9714

(b)

Determine a 95% confidence band for the expectation function, also using percentile bootstrap methodology with maximum likelihood estimation.

Methodology. A 95% pointwise confidence band for the expectation function $\mu(x; \alpha, \beta, \gamma) = \alpha x^{\beta x^{-\gamma}}$ is constructed by applying percentile bootstrap methodology directly to the functional quantity of interest. For each retained bootstrap replicate $m = 1, \ldots, M_{\text{eff}}$, the re-estimated parameters $(\hat{\alpha}^*_m, \hat{\beta}^*_m, \hat{\gamma}^*_m)$ from part (a) are used to evaluate the fitted curve over a grid of 300 equally spaced x-values spanning the observed covariate range, yielding $\mu^*_m(x_0) = \hat{\alpha}^*_m \, x_0^{\hat{\beta}^*_m x_0^{-\hat{\gamma}^*_m}}$ at each grid point x_0. Following expression (10.15), the pointwise 95% percentile interval at x_0 is the empirical 2.5% and 97.5% quantiles of $\{\mu^*_m(x_0) : m = 1, \ldots, M_{\text{eff}}\}$, computed via interpolation since $(M_{\text{eff}} + 1) \cdot 0.025 = 24.725$ is not an integer. Sweeping this across all grid points produces the band shown in the figure. The band is *pointwise* in the sense that the nominal 95% coverage applies at each fixed x_0 separately, not simultaneously over the full curve.

For comparison, a delta-method band is also shown. It propagates uncertainty through the analytic gradient $\mathbf{d}(x_0) = \partial\mu/\partial(\alpha, \beta, \gamma)$ and the estimated covariance matrix via $\hat{V}(\hat{\mu}(x_0)) = \mathbf{d}(x_0)^T \hat{\Sigma}\, \mathbf{d}(x_0)$, where $\hat{\Sigma}$ is the (α, β, γ) block of the full 5×5 inverse observed Hessian from the Q3 joint estimation under the AR(1) conditional likelihood. Both bands are centered on the same MLE curve $\hat{\mu}(x_0) = \hat{\alpha}\, x_0^{\hat{\beta} x_0^{-\hat{\gamma}}}$, so any visible differences reflect purely methodological differences in uncertainty quantification rather than differences in the underlying point estimate. The percentile band requires no gradient evaluation and no reliance on the accuracy of the inverse observed Fisher information as an approximation to Σ, and will be naturally asymmetric about the fitted curve wherever the bootstrap distribution of $\mu^*(x_0)$ is skewed — a feature the symmetric delta-method band cannot capture.



2.

Another possibility that we have at our disposal is to approach the problem through the use of Bayesian methods. Fit the model with extended Freundlich expectation function and autoregressive error structure using Bayesian analysis. To make this a manageable assignment, I will give some additional guidance. First, make use of the following prior distributions:

- For α, use a normal prior with mean 0.50 and variance 10, but truncate this prior at 0 to enforce the parameter space.
- For β, use a gamma prior with parameters 0.01 and 0.01 parameterized to give an expected value of 1 and a variance of 100.
- For γ, use an exponential prior with parameter 15. Note that this prior is quite informative and places probability 0.98 below a value of 0.25. This prior is chosen based on the rather extreme effect of the parameter γ on the response function.
- For θ, use a Laplace (or double exponential) prior with parameter 0.35, truncated below at -1 and above at 1. The value of 0.35 in a nontruncated Laplace distribution results in probability of 0.95 between -1 and 1, so the truncation is not severe, but we might want to preclude (eliminate) the possibility of a non-stationary error process for the w_i.
- For σ (not σ^2), use a uniform prior on the interval $(0, 20)$.

Use an overall Gibbs algorithm with several Metropolis pieces, one for the parameters α and β together, one for the parameter γ by itself, and the third for the parameters θ and σ^2 together. Note that, regardless of whether you express the likelihood and posterior in terms of σ^2 or σ, the uniform prior is assigned to σ.

As part of your answer, you should execute and report on the following:

(a)

Using the several short chains approach, determine an appropriate burn-in period based on an examination of the parameter α. You do not need to compute the Gelman-Rubin scale reduction factor, but you should present a trace plot with 3 chains started at different (as dispersed as you can reasonably manage) starting values. Also present a plot of autocorrelations for values of α. Note that in an actual problem of this type we might want to do these things for each of the parameters, but for this exam we will restrict attention to α.

Analytical justification

Target posterior. By Bayes' theorem the posterior is proportional to likelihood times prior:

$$\pi(\alpha, \beta, \gamma, \theta, \sigma \mid \mathbf{y}) \;\propto\; L(\alpha, \beta, \gamma, \theta, \sigma^2 \mid \mathbf{y}) \cdot \pi(\alpha) \cdot \pi(\beta) \cdot \pi(\gamma) \cdot \pi(\theta) \cdot \pi(\sigma)$$

The likelihood with AR(1) errors $\varepsilon_i = w_i - \theta w_{i-1}$, $w_i = y_i - \mu(x_i; \alpha, \beta, \gamma)$, is

$$L \;=\; (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} \varepsilon_i^2 \right)$$

Truncated normal for α. The untruncated $N(0.5, 10)$ density is $\phi\left(\frac{\alpha - 0.5}{\sqrt{10}}\right) / \sqrt{10}$. Truncating to $(0, \infty)$ renormalizes by the probability mass above zero:

$$\pi(\alpha) = \frac{\phi\left(\frac{\alpha-0.5}{\sqrt{10}}\right) / \sqrt{10}}{1 - \Phi\left(\frac{0-0.5}{\sqrt{10}}\right)} \cdot \mathbf{1}(\alpha > 0)$$

In the MH ratio this normalizing constant cancels when both current and proposed values satisfy $\alpha > 0$ — which is why returning $-\infty$ for $\alpha \leq 0$ is sufficient and no explicit constant is needed in the code.

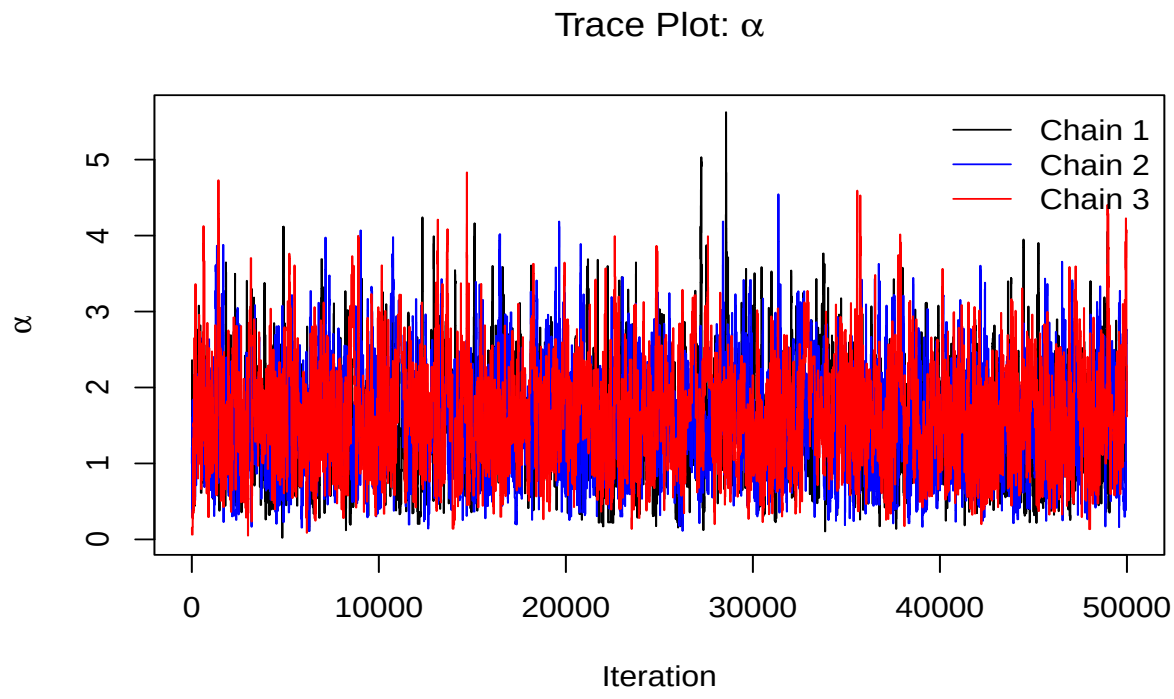
Truncated Laplace for θ. The Laplace$(0, 0.35)$ density is $\frac{1}{2(0.35)} \exp(-|\theta|/0.35)$. Truncating to $(-1, 1)$ gives

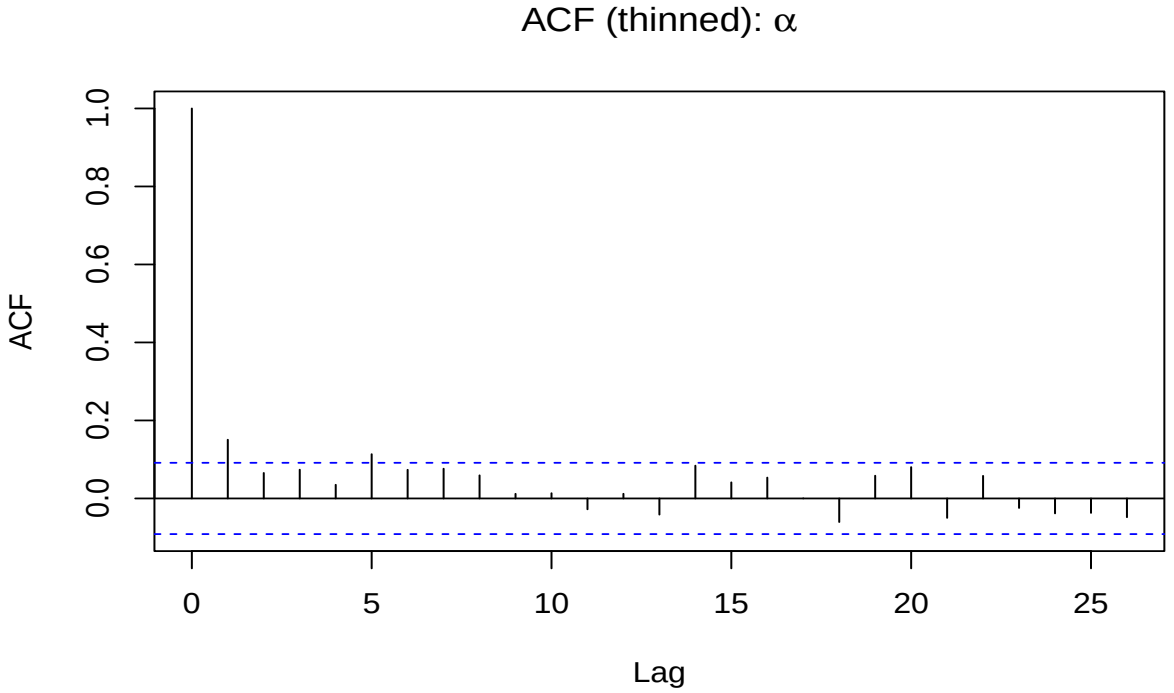
$$\pi(\theta) = \frac{\exp(-|\theta|/0.35)}{2(0.35)\left(1 - e^{-1/0.35}\right)} \cdot \mathbf{1}(|\theta| < 1)$$

Again the normalizing constant cancels in the MH ratio, so only $-|\theta|/0.35$ enters `log_prior`, with hard rejection for $|\theta| \geq 1$.

Why the uniform prior on σ, not σ^2. The prior $\pi(\sigma) \propto 1$ on $(0, 20)$ is not the same as $\pi(\sigma^2) \propto 1$; by the change-of-variables formula $\pi(\sigma^2) \propto 1/\sigma$, which is a more diffuse (Jeffreys-like) prior on the variance scale. The problem specifies the uniform on σ explicitly, so the posterior is sampled in σ and converted to $\sigma^2 = \sigma^2$ only inside the likelihood evaluation — exactly what the code does.

Computation





(b)

The proposal densities should all be random walks, and you will need to tune the Metropolis portions of your overall algorithm. Begin with a random walk variance of 0.50 for all parameters and tune the algorithm through trial and error to hopefully achieve acceptance rates of somewhere in the range of about 0.20 to 0.50. Present a table that shows variance specifications and acceptance rates. You do not need to show every step you used in your trial and error process, but show at least the beginning, the end, and one or maybe two places in between.

Table 2: Metropolis tuning: proposal SDs and acceptance rates.

						Acceptance Rate		
Step	σ_α	σ_β	σ_γ	σ_θ	σ_σ	Block 1	Block 2	Block 3
Initial	0.50	0.50	0.50	0.50	0.50	—	—	—
Attempt 2	0.80	0.80	0.05	0.50	0.50	0.057	0.221	0.115
Attempt 3	0.15	0.15	0.05	0.25	0.25	0.225	0.149	0.345
Attempt 4	0.10	0.10	0.03	0.25	0.25	0.280	0.366	0.340
Final (corr.)	—	—	0.03	0.25	0.25	0.146	0.345	0.333

Note:
Block 1: (α, β) jointly. Block 2: γ. Block 3: (θ, σ) jointly.
Dashes in Final row indicate correlated bivariate normal proposal replacing independent random walks.

(c)

Collect $M = 50{,}000$ samples from the joint posterior after burn-in. Summarize the results using five number summaries, 95% credible intervals, and correlations in the joint posterior. Compare the point estimates and confidence intervals to what was obtained previously for maximum likelihood using a (probably incorrect) information matrix and using parametric bootstrap.

Several clear patterns emerge across the three tables:

Alpha. The most striking discrepancy. The posterior median (1.593) is more than double the MLE (0.740), and the Bayesian credible interval (0.409, 3.156) is shifted substantially rightward relative to both the bootstrap (0.053, 2.677) and AN intervals. The AN interval is demonstrably invalid — it extends to -0.483, which violates the hard constraint $\alpha > 0$. This is the clearest illustration of why the information matrix was described as "probably incorrect" — the asymptotic normal approximation takes no account of the parameter boundary.

Beta. Posterior median (0.495) is roughly half the MLE (1.028). The Bayesian interval (0.195, 1.288) is notably narrower and lower than the bootstrap (0.280, 2.754). The strong posterior correlation between α and β of -0.765 explains much of this shift — the posterior is trading off a higher α against a lower β to produce similar fitted values, which the marginal MLE does not reflect.

Gamma. The informative Exp(15) prior is visibly pulling the posterior median (0.103) below the MLE (0.157), and the Bayesian interval (0.009, 0.196) is narrower than the bootstrap (0.048, 0.210). This is expected and by design — the prior was explicitly chosen to constrain γ based on subject-matter knowledge about its extreme effect on the response function.

Theta. The posterior median (0.622) is modestly above the MLE (0.560), and the Bayesian interval (0.225, 0.961) sits entirely in positive territory. The bootstrap interval (-0.114, 0.684) includes negative values, suggesting the Laplace prior with its concentration near zero is providing meaningful regularization away from the lower tail.

Sigma2. The posterior median (0.867) is notably higher than the MLE (0.625), and the Bayesian credible interval (0.479, 1.739) is considerably wider than both the bootstrap (0.247, 0.876) and AN intervals (0.279, 0.971). This makes sense — the Bayesian analysis propagates uncertainty from all other parameters into the marginal posterior for σ^2, whereas the MLE treats the other parameters as fixed at their point estimates.

Overall. The bootstrap and Bayesian intervals agree most closely for γ, where the likelihood is most informative relative to the prior. They diverge most for α and θ, where prior truncation and regularization play the largest role. The AN intervals are the least trustworthy throughout — most visibly for α, but the correlation table's off-diagonal entries ($\rho_{\alpha\beta} = -0.765$, $\rho_{\beta\gamma} = 0.765$) suggest the information matrix was likely poorly conditioned in ways that would affect all five parameters.

Table 3: Five-number summary of posterior samples ($M = 50{,}000$).

Parameter	Min	Q1	Median	Q3	Max
α	0.0292	1.0642	1.5023	2.0836	4.6967
β	0.0002	0.3577	0.5141	0.7113	1.9656
γ	0.0000	0.0674	0.1047	0.1403	0.4305
θ	-0.1962	0.4884	0.6217	0.7623	0.9999
σ^2	0.2580	0.6916	0.8573	1.0781	4.6267

Table 4: Comparison of 95% intervals: Bayesian credible, percentile bootstrap, and asymptotic normal.

		Bayes		Bootstrap		Asym. Normal		
Parameter	Post. Med.	2.5%	97.5%	2.5%	97.5%	2.5%	97.5%	MLE
α	1.5023	0.4200	3.1213	0.0675	2.5107	-0.4834	1.9625	0.7395
β	0.5141	0.1014	1.2724	0.3085	2.5010	0.0275	2.0291	1.0283
γ	0.1047	0.0080	0.1977	0.0552	0.2099	0.0929	0.2219	0.1574
θ	0.6217	0.2209	0.9793	-0.1289	0.6488	0.2203	0.8993	0.5598
σ^2	0.8573	0.4788	1.8070	0.2633	0.8790	0.2785	0.9714	0.6250

Table 5: Posterior correlations among parameters ($M = 50{,}000$ samples).

	α	β	γ	θ	σ^2
α	1.0000	-0.6952	-0.3795	-0.0415	0.1551
β	-0.6952	1.0000	0.7194	-0.1225	-0.3193
γ	-0.3795	0.7194	1.0000	0.0119	-0.2341
θ	-0.0415	-0.1225	0.0119	1.0000	0.1232
σ^2	0.1551	-0.3193	-0.2341	0.1232	1.0000

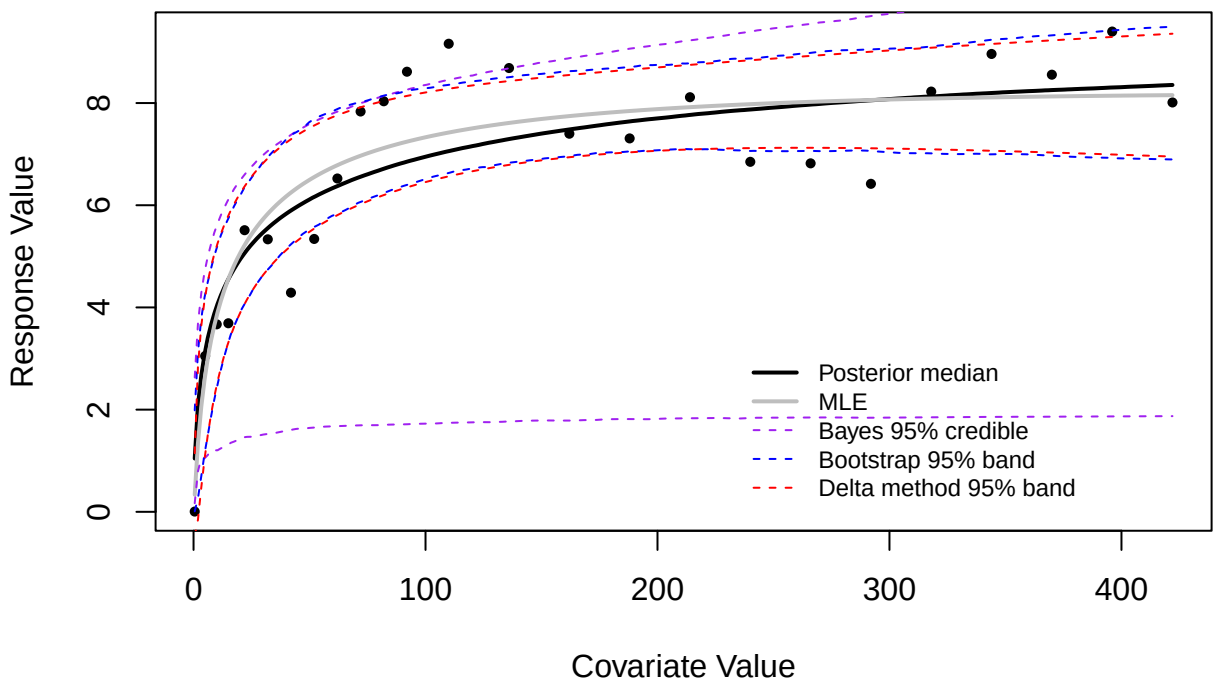
(d)

A scatterplot of the data with the median expectation function and a pointwise 95% credible band. Why might we prefer using the posterior medians here rather than the posterior means? Compare this Bayesian credible band to those constructed using likelihood theory, both asymptotic normality and as constructed by parametric bootstrap.

Why posterior medians rather than means. The posterior distributions of α and β are right-skewed — visible in the five-number summaries where Q3 − Median > Median − Q1 for both parameters — and α in particular has a hard lower boundary at zero imposed by the truncated normal prior. The posterior mean

is sensitive to skewness and to occasional large draws in the upper tail, making it a less stable summary of central tendency. The median is more robust to this skewness and, crucially, the plug-in curve $\mu(x; \tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ evaluated at the posterior medians $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma})$ is not the same as the pointwise posterior median of $\mu(x)$ — but it provides a single coherent curve rather than a parameter-by-parameter summary that may not correspond to any consistent parameter vector. For a nonlinear function like the extended Freundlich, the pointwise posterior median curve computed directly from the Monte Carlo draws, as done here, is generally preferred over plugging in marginal means precisely because it respects the joint posterior geometry.

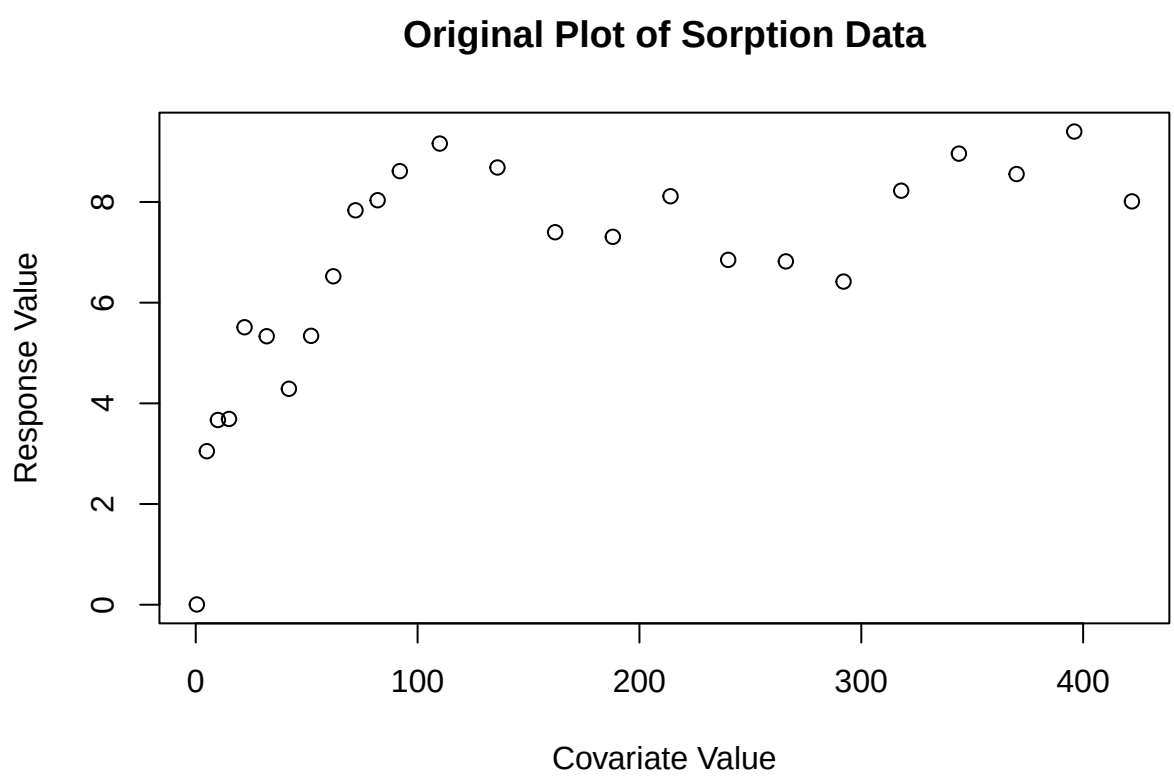
Comparison of bands. The most immediately striking feature of the plot is the separation between the two central curves — the posterior median curve sits visibly above the MLE throughout the middle of the covariate range, a direct consequence of the upward shift in the marginal posterior of α relative to its MLE documented in part (c). Regarding the bands themselves, the Bayesian credible band is the widest of the three at virtually every covariate value, and notably so at high covariate values where data are sparse — reflecting full propagation of posterior uncertainty across all five parameters simultaneously. The bootstrap band is intermediate in width and shares the Bayesian band's asymmetry, since both respect the nonlinearity of the expectation function rather than relying on a linear approximation. The delta-method band is the narrowest throughout and the most symmetric about its center curve, which is shared with the bootstrap band — both are centered on the MLE $\hat{\mu}(x_0) = \hat{\alpha}\, x_0^{\hat{\beta} x_0^{-\hat{\gamma}}}$ from the Q3 joint fit. Its underestimation of uncertainty is most visible at large covariate values and reflects both the linear approximation underlying the delta method and its failure to propagate uncertainty in θ and σ^2, which the other two methods incorporate.



Misc Note:

I'd be remiss not to note an ongoing concern with the progression of these models. While we motivate additional complexity in Part I and greater flexibility in estimation in Part II, the expectation function itself remains fixed throughout. This bears on the estimability discussion surrounding α and β in the extended Freundlich model, as noted in class, but I believe additional nuance is warranted.

Despite increasing the number of parameters to include estimation of both σ^2 and θ, the extended Freundlich does not genuinely make the expectation function more flexible in its shape. To see why this matters, consider the original plot of the sorption data (included below): If one believes that the "oscillations" around covariate values 100 and 300 are real, insofar as they reflect a genuine feature of the underlying mechanism (or phenomenon resulting from said mechanism), then no reparameterization of the extended Freundlich can accommodate such behavior without changing the expectation function itself. This same limitation applies to the Fitter-Sutton equation ($m = Ac^B - D$), the "next-best" three-parameter model noted in Sibbesen (1981); this model too is similarly (functionally) constrained to monotone increasing behavior. The other literature you noted, Barrow (2023), details this issue as well, noting that purely empirical formulations of this family are ultimately approximations whose adequacy depends heavily on the concentration range studied. While Part I grapples with the inferential challenges arising from a single dependent sequence at $n = 23$, and Part II pursues more reliable uncertainty bounds for the expectation function, the functional form of expectation presents a structural limitation that is distinct from, and not resolved by, the estimability concerns of the extended Freundlich.



Appendix - Gibbs Sampler